

Tarra Sathwik

Email: 23ec01047@iitbbs.ac.in

LinkedIn: [Sathwik](#)

GitHub: [Sathwik182079](#)

LeetCode: [sathwik](#)

Education

Indian Institute of Technology, Bhubaneswar

CGPA: 7.26

B.Tech., Electronics and Computer Science Engineering

Aug. 2023 - Present

Sri Chaitanya College

Percentage: 94%

Senior School Certificate Examination

June 2021 - March 2023

Dr KKR Gowtham High School

Percentage: 99.9%

Secondary School Examination

June 2016 - March 2021

Internship

Research Intern at IIT Bhubaneswar

(May 2026 - July 2026)

Under Dr. Sandeep Kumar Samal, Department of Electronics and Communication

- Gained hands-on experience with the digital ASIC design flow using Cadence Innovus, Voltus, Synopsys PrimeTime, and HSPICE.
- Performed floorplanning, placement, clock tree synthesis (CTS), routing, and power integrity analysis, Conducted Static Timing Analysis (STA)
- Simulated CMOS circuits using HSPICE and worked in a Linux-based EDA environment for design, analysis, and debugging.

Research Intern at IIT Bhubaneswar

(May 2025 - July 2025)

Under Dr. Devashree Tripathy, Department of Computer Science

- Gained in-depth expertise in GPU architecture and CUDA's software programming interface, and leveraged new features of the latest H100 GPUs to optimize and accelerate existing applications.
- Developed and executed CUDA benchmarking scripts for NVIDIA H100 GPUs to analyze and optimize parallel computing performance.

Projects

5-Stage Pipelined RISC-V Processor

Verilog HDL / Vivado

- Designed a 32-bit 5-stage pipelined RISC-V processor with hazard detection, data forwarding, and branch handling.
- Implemented IF/ID/EX/MEM/WB pipeline, ALU, Register File, Control Unit, Instruction/Data Memory, and pipeline registers in Verilog.

I²C Master-Slave Controller

Verilog / Vivado

- Designed RTL for I²C Master and Slave controllers using Verilog HDL. Implemented START, STOP, repeated START, ACK/NACK, open-drain SDA, and multi-byte read/write transactions.
- Developed FSM-based protocol control and clock divider for SCL generation.

Course Projects

bfloat-16

Verilog

- This is a project that adds or subtracts two bfloat-16 numbers, implemented on FPGA basys3 board.

Asynchronous FIFO

Verilog

- This project implements an Asynchronous FIFO (First-In First-Out) using Verilog HDL.

Key Skills

- **Programming Languages:** C, C++, Python, CUDA, Verilog HDL, Matlab
- **Hardware Design:** RTL Design, Digital Logic Design, Computer Architecture, GPU architecture, RISC-V Processor Design, Parallel Programming
- **EDA Tools:** Xilinx Vivado, Cadence Virtuoso, Cadence Innovus, Cadence Voltus, Synopsys PrimeTime, Synopsys HSPICE
- **Soft Skills:** Problem solving, Teamwork, Communication, Time management

Position Of Responsibility

- Student Guide, Counseling Service Team- IIT Bhubaneswar
- Assistant Coordinator, Counseling Service Team – IIT Bhubaneswar
- Sponsors Core Head member | Pravaah, Annual Cultural Technopreneurship Fest of IIT Bhubaneswar
- Sponsors Core Team member | Pravaah, Annual Cultural Technopreneurship Fest of IIT Bhubaneswar
- Sponsors team Associate | Wissenaire, Techno-Management Fest: